A Solutions to the exercises

A.1 Chapter 2: Gaussian processes as distributions over functions

In this exercise, we will train a Gaussian process on a real-world data set that we saw in chapter 1, which is shown again in Table A.1.

Each data point (row) corresponds an alloy (a kind of metal) created by mixing lead (Pb), tin (Sn), germanium (Ge), and manganese (Mn)—these are called the *parent compounds*—at different ratios. The features are contained in the first four columns, which are the percentages of the parent compounds. The prediction target, mixing temperature, is in the last column, denoting the lowest temperature at which an alloy can form. The task is to predict mixing temperature given the compositional percentages of an alloy.

Table A.1 Data from a materials discovery task. The features are the structure of a material expressed in percentages of parent compounds, and the prediction target is the mixing temperature.

% of Pb	% of Sn	% of Ge	% of Mn	Mixing temp.
0.50	0.50	0.00	0.00	192.08
0.33	0.33	0.33	0.00	258.30
0.00	0.50	0.50	0.00	187.24
0.00	0.33	0.33	0.33	188.54

There are multiple steps:

1. Create the four-dimensional dataset included in Table A.1.
2. Normalize the fifth column by subtracting the mean from all values and dividing the results by their variance.
3. Treat the first four columns as features and the fifth as labels. Train a Gaussian process on this data. You can reuse the GP model class we implemented in the chapter.
4. Create a test data set containing compositions having zero percentage of germanium and manganese. In other words, the test set is a grid over the unit square whose axes are

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